

VLM-Guided Detection and Rematching for Private Object Localization

Heeseung Cho¹ Yuna Park¹ Esther Kim² Jiho Kim² Hyojoo Kim²
Christian Wallraven^{1,3} Junhyoung Oh^{2*}

¹Department of Artificial Intelligence, Korea University

²Division of Information Security, Seoul Women’s University

³Department of Brain and Cognitive Engineering, Korea University

Abstract

We present a training-free framework for few-shot private object localization based on coarse-to-fine semantic alignment. We retain an off-the-shelf open-vocabulary detector(OVD) and align it through semantic guidance from vision-language model(VLM). Given any query image, our approach operates in two phases: (i) a VLM generates category-aware cues and identifies a PII-type supercategory to guide Grounding DINO toward privacy-relevant regions, and (ii) the same VLM jointly re-evaluates all detected candidates on the full image, assigning categories within a supercategory-restricted candidate space and rejecting false positives. SAM produces the final segmentation masks. Experiments on the Biv-Priv-Seg dataset demonstrate strong performance without any model fine-tuning or support augmentation, surpassing the 2025 top-performing method LLM2Seg by +3.64 bbox mAP and +3.68 segmentation mAP. Our code is available at https://github.com/Heeseung-Cho/semantic_vlm_privacy.

1. Introduction and Problem Statement

The VizWiz Few-Shot Object Localization (FSOL) challenge [3] requires localizing 16 private object categories in images captured by blind and low vision (BLV) users. Participants receive a support set of 16 labeled images (one per category) and a query set of 1,056 unlabeled images, and must predict bounding boxes and segmentation masks without external training data. Prior study [3] shows that open-vocabulary detector(OVD) plateau at AP₅₀ 60% even with oracle category names, while traditional few-shot detectors yield only 12.7 mAP, indicating a ceiling in detector-centric approaches.

Instead of fine-tuning the detector, we retain an off-the-shelf OVD and maximize its coverage through VLM-generated semantic cues. We propose a coarse-to-fine framework: a vision-language model(VLM) first generates

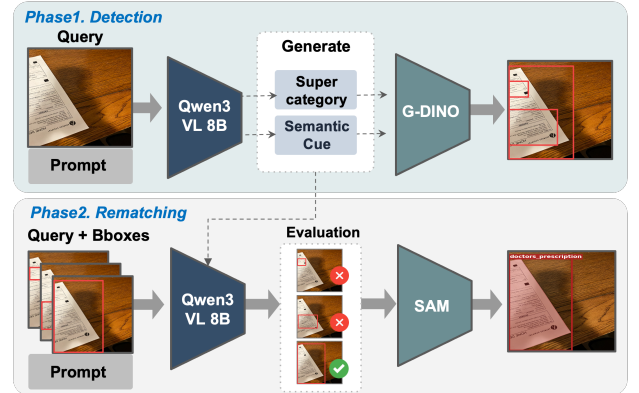


Figure 1. Overview of our proposed method.

category-aware cues that guide the detector toward privacy-relevant regions, then jointly re-evaluates all detected candidates to correct categories within a restricted label space.

2. Method

Figure 1 shows our framework performing concept alignment in two phases around an OVD (Grounding DINO). In Phase 1 (*Detection*), a VLM (Qwen3-VL-8B) routes the query image to a category and passes it as a text prompt to the detector, which generates candidate bounding boxes based on the VLM-inferred category. In Phase 2 (*Rematching*), the full query image is annotated with the proposed bounding boxes and presented to the VLM together with the Phase-1 outputs, a scene caption, and per-box detector context. Given this combined information, the VLM selects valid candidates by accepting or rejecting each box and assigns a final category from the candidate space. Accepted predictions are passed to SAM for segmentation.

We introduce two components to enhance this baseline. (i) *Semantic Cue*: the VLM generates detector-friendly noun phrases that describe the target beyond its category name, enriching the Phase-1 detection prompt. (ii) *Super-*

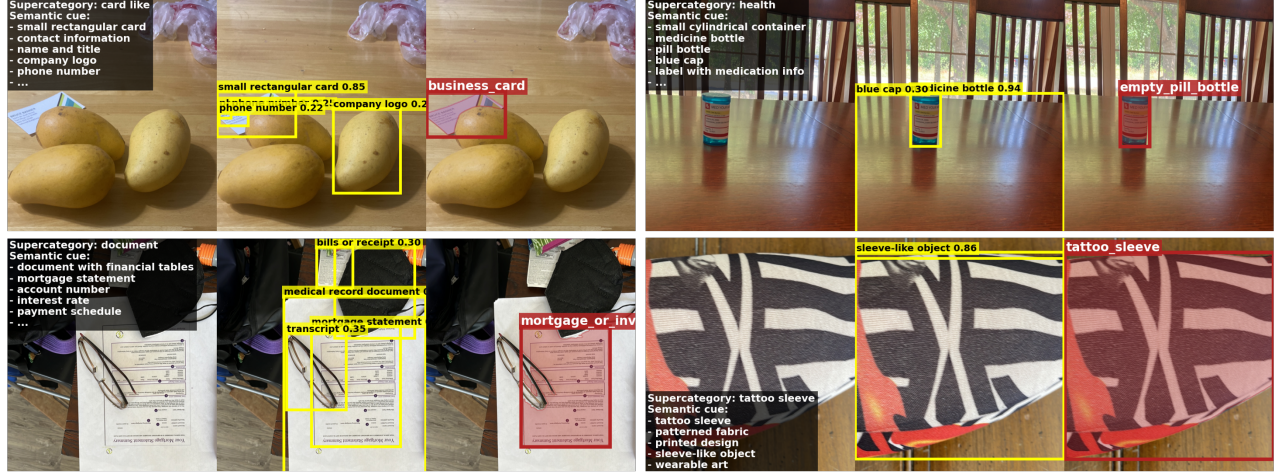


Figure 2. Pipeline examples for each supercategories. (Left) Phase-1 VLM output. (Middle) Candidates produced from Phase-1 prompt. (Right) Final predictions after Phase-2, which rejects false positives and assigns the correct category.

	Ablation	bbox		segmentation	
		mAP	AP ₅₀	mAP	AP ₅₀
LLM2Seg [4]		61.63	64.95	60.14	63.49
Ours	–	56.34	53.24	54.69	52.16
	+ Cue	63.44	60.25	61.85	58.99
	+ SC	64.87	61.29	63.19	60.17
	+ Cue + SC	65.27	61.92	63.82	61.00

Table 1. Ablation and comparison on the FSOL task.

category (SC) Expansion: drawing on the PII taxonomy in [2], we define four supercategory groups—*document* (8 categories), *health* (5), *card* (2), and *tattoo* (1)—and broaden the candidate space from the single predicted category to all categories within the predicted group. SC expansion benefits both phases: the detector receives a wider set of search targets, and the VLM can reassign categories that the initial prediction may have missed.

3. Results and Discussion

As shown in Tab. 1, our method achieves 65.27 bbox mAP and 63.82 segmentation mAP without any fine-tuning, outperforming LLM2Seg [4] by +3.64 and +3.68 respectively. Our baseline—per-image VLM evaluation with fixed category names and no SC expansion—achieves 56.34 bbox mAP. Semantic cue generation adds +7.10 and SC expansion adds +8.53 independently; combining both yields the best result at +8.93 over the baseline. Fig. 2 illustrates the full pipeline on examples from different SC. These results point to the followings: First, the detector ceiling in [3] reflects a structural limitation of OVD; VLM-generated cues expand coverage without modifying the detector. Second,

jointly rematching candidates per image [1] enables false positive rejection and category reassignment.

We demonstrate that coarse-to-fine semantic alignment around an off-the-shelf detector outperforms detector adaptation for BLV private object localization, offering a practical direction for few-shot localization in privacy-sensitive domains.

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References

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